CONTROL

Motor speed → thrust → torque → Angular acceleration

τ = Torque

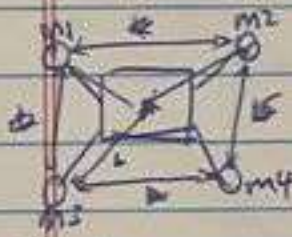
τ_{roll} = Force x distance

Scenario

If a drone wants to turn clockwise,

The first 2 motors on the left produce more thrust and the two motors on the right produce less thrust. There is an imbalance in forces causing a rotation.

- Torque τ = distance x Force x $\sin\theta$
- τ = distance from the center of mass to the motor x ~~Force~~ Thrust



Each motor produce a thrust

the thrust

Increasing m_1 and m_3 while decreasing the thrust in m_2 and m_4 produces a net roll torque

$$Thrust(m_1) + Thrust(m_3) > Thrust(m_2) + Thrust(m_4)$$

Because the forces are acting away from each other, they create a rotational effect.

Torque = distance from the center of mass to the motor X Thrust

$\rightarrow \quad \tau = L\bar{T}$

Net roll torque

$$L(\bar{T}_1 + \bar{T}_3) > L(\bar{T}_2 + \bar{T}_4)$$

$$\tau_\phi = L(\bar{T}_1 + \bar{T}_3) - L(\bar{T}_2 + \bar{T}_4) \quad \text{—②}$$

angular acceleration, $\alpha = \frac{d\omega}{dt}$ $\frac{\text{velocity}}{\text{time}}$

Newton's second law,

Torque = Moment of inertia × angular acceleration

$\tau = J\alpha$ — (2)

(1) = (2)

$L(\bar{T}_1 + \bar{T}_3 - \bar{T}_2 - \bar{T}_4) = J\alpha$

$\alpha = \frac{L}{J}(\bar{T}_1 + \bar{T}_3 - \bar{T}_2 - \bar{T}_4)$

The STM32 will not command thrust directly
The STM32 will command something like,

PWM → ESC → motor → propeller

motor speed → thrust
motor speed produces thrust.

~~motor speed → thrust → torque → angular acc~~

STRATEGY

→ Physics : Thrust → Torque → Angular acceleration

→ Mathematical layer : $G(s) = \frac{\text{output}(s)}{\text{input}(s)}$

→ Digital model : $G(s) \rightarrow G(z)$, discretize the plant and;
$G(z) \rightarrow D(z)$

→ Simulation

→ Firmware
$D(z) \rightarrow$ Difference equation → C++ → STM32

Moment of Inertia

$J = mr^2$

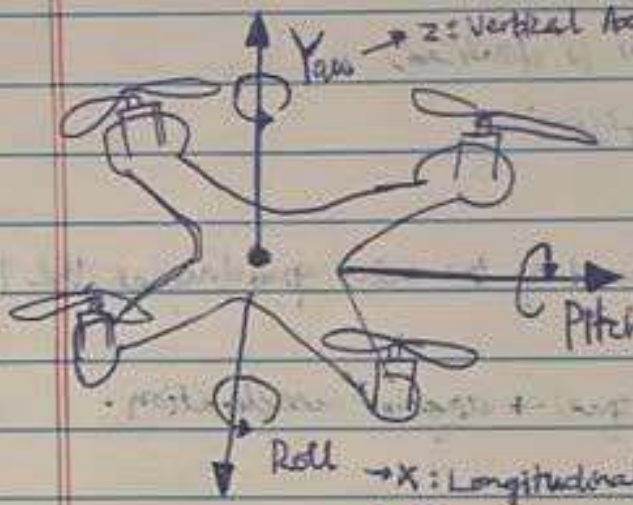
J_{roll} = J of all the components

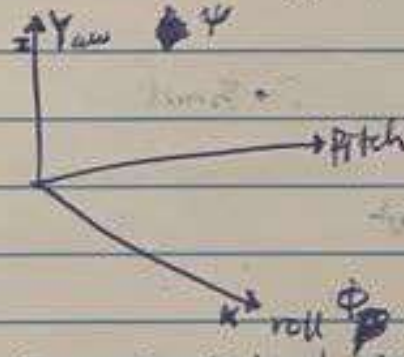
$J_{roll} = J_{frame} + J_{motor} + J_{battery} + J_{esc} + J_{propeller} + \cdots$

Roll → Either tilting left or tilting right.

Pitch → Either tilt in the forward or the rear

Yaw → Turn right or turn left of the whole structure.





- Roll = From front to back
  Rotation around it makes the drone tilt left/right
- Pitch = From left to right
  Rotation around it makes the drone tilt forward/backwards
- Yaw, z-axis = Vertical through the drone.
  Rotation around it makes the drone turn left / right

THRUST

$\bar{T} = K_T \omega^2$

K_T = thrust coefficient

ω = motor / propeller speed.

If $\omega = 100$:

$\bar{T}_1 = K_T (100)^2$ If ω is doubled,

$\bar{T}_2 = K_T (200)^2 \Rightarrow \bar{T}_2 = K_T (100)^2 4$

$\bar{T}_2 = 4\bar{T}_1$

Therefore, doubling the speed of the motor quadruples the thrust.

For each motor;

$\bar{T}_1 = K_T \omega_1^2$

$\bar{T}_2 = K_T \omega_2^2$

$\bar{T}_3 = K_T \omega_3^2$

$\bar{T}_4 = K_T \omega_4^2$

Speed → Thrust → Torque → angular acceleration.

$\bar{T} = J\alpha$

$\alpha = \frac{\bar{T}}{J}$ $\quad \bar{T} = L\bar{T}$ $\quad \bar{T}$ → Thrust

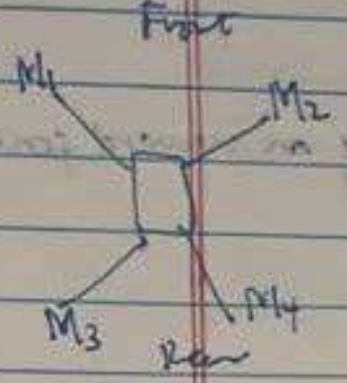
$\alpha = \frac{L\bar{T}}{J}$ $\quad \bar{T} = K_T \omega^2$

$\alpha = \frac{L K_T \omega^2}{J}$

If we wanna roll it; one side's speed increases than the other.

$T_{roll} = L K_T (\omega_1^2 + \omega_3^2 - \omega_2^2 - \omega_4^2)$

Therefore: $\alpha_{roll} = \frac{L K_T}{J_{roll}} \left[\omega_1^2 + \omega_3^2 - \omega_2^2 - \omega_4^2 \right]$



If the rear motors produce more thrust than the front,

$\bar{\tau}_{pitch} = L(\bar{T}_3 + \bar{T}_4 - \bar{T}_1 - \bar{T}_2)$

$\bar{T}_i = K_T \omega_i^2$

~~$\bar{\tau}_{pitch} =$~~

$\theta_{pitch} = \frac{\tau_{pitch}}{J_{pitch}}$

$\theta_{pitch} = \frac{LK_T}{J_{pitch}} \left[\omega_3^2 + \omega_4^2 - \omega_1^2 - \omega_2^2\right]$

Yaw

For Yaw we increase M1 and M4 and decrease M2 and M3

$\tau_{yaw} = L(\bar{T}_1 + \bar{T}_4 - \bar{T}_2 - \bar{T}_3)$

$\bar{T}_i = K_Q \omega_i^2$

$\theta_{yaw} = \frac{\tau_{yaw}}{J_{yaw}} = \frac{LK_Q}{J_{yaw}} \left[\omega_1^2 + \omega_4^2 - \omega_2^2 - \omega_3^2\right]$

All rotational parts have been derived above now !!

$\bar{T}_1 = K_T \omega_1^2$

$\bar{T}_2 = K_T \omega_2^2$ Thrust is simply an upward force

$\bar{T}_3 = K_T \omega_3^2$

$\bar{T}_4 = K_T \omega_4^2$

Total thrust, $\bar{T} = \bar{T}_1 + \bar{T}_2 + \bar{T}_3 + \bar{T}_4$

$\bar{T} = K_T [\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2]$

$\tau_{roll} = LK_T [\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2]$

$\tau_{pitch} = LK_T [\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2]$

$\tau_{yaw} = LK_Q [\omega_1^2 - \omega_2^2 - \omega_3^2 + \omega_4^2]$

$$\begin{bmatrix} \bar{T} \\ \tau_{roll} \\ \tau_{pitch} \\ \tau_{yaw} \end{bmatrix} = \begin{bmatrix} K_T & K_T & K_T & K_T \\ LK_T & -LK_T & LK_T & -LK_T \\ LK_T & LK_T & -LK_T & -LK_T \\ K_Q & -K_Q & -K_Q & K_Q \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix}$$

→ drone geometry + motor arrangement

The system we want

Desired angle → controller → required torque → motor commands

plant → Actual angle

Sensor ←

To make the drone hover,

$\bar{T} = mg$, $\tau_{roll} = \tau_{pitch} = \tau_{yaw} = 0$

ϕ

θ

ψ

The equations or the matrix, will be used to calculate and determine how the drone should be positioned and also move.

Now; we determine the various torque angles.

$T_{roll} = J_{roll} \alpha_{roll}$

$\alpha_{roll} = \ddot{\phi}$ $\quad \ddot{\phi}$ = angular acceleration

$T_{roll} = J_{roll} \ddot{\phi}$

$\ddot{\phi} = \frac{T_{roll}}{J_{roll}}$

Change the acceleration into angle.

$\alpha = \frac{dv}{dt}$; $v = \frac{dx}{dt}$; $\alpha = \frac{d^2x}{dt^2}$

$\Rightarrow \alpha = \frac{d\omega}{dt}$; $\omega = \frac{d\phi}{dt}$; $\alpha = \frac{d^2\phi}{dt^2} \Rightarrow \ddot{\phi} = \frac{d^2\phi}{dt^2}$

$T_{roll} = J_{roll} \ddot{\phi}$

$$\boxed{T_{roll} = \frac{d^2\phi}{dt^2} J_{roll}}$$ → use this to derive the transfer functions.

$\Rightarrow T_{roll}(s) = J_{roll} s^2 \phi(s)$ $\quad \frac{\text{output}}{\text{input}}$

$$\boxed{\frac{\phi(s)}{T_{roll}(s)} = \frac{1}{s^2 J_{roll}}}$$

This is just for the roll torque.

$H(s) = \frac{1}{s^2 J_{roll}}$; If you apply a roll torque to the drone, the drone's roll angle changes according to the rotational inertia.

Roll torque:

$$\frac{\phi(s)}{T_{roll}(s)} = \frac{1}{s^2 J_{roll}}$$

Pitch torque:

$$\frac{\theta(s)}{T_{Pitch}(s)} = \frac{1}{s^2 J_{pitch}}$$

Yaw torque:

$$\frac{\psi(s)}{T_{yaw}(s)} = \frac{1}{s^2 J_{yaw}}$$

So; the T_{roll} (roll torque) is produced from motor speeds.
$\phi(s)$ (roll angle) is produced from a roll torque.

$$\phi(s) = \frac{1}{s^2 J_{roll}} T_{roll}(s)$$

$$T_{roll} = LK_T(-\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$$

Therefore: $\phi(s) = \frac{1}{s^2 J_{roll}} LK_T(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$

pitch

$$\theta(s) = \frac{1}{s^2 J_{pitch}} \bar{T}_{pitch}(s)$$

$$T_{pitch} = LK_T(\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2)$$

$$\theta(s) = \frac{1}{s^2 J_{pitch}} LK_T[\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2]$$

Yaw

$$\psi(s) = \frac{1}{s^2 J_{yaw}} \bar{T}_{yaw}(s)$$

$$T_{yaw} = \frac{1}{s^2 J_{yaw}} LK_Q[\omega_1^2 - \omega_2^2 - \omega_3^2 + \omega_4^2]$$